

Exercises in Marine Ecological Genetics

09. DNA barcoding

- DNA barcodes
- Barcode of Life Data system (BOLD)
- p -distances
- Barcoding gap



Martin Helmkamp

<https://github.com/mhelmkampf/meg26>

Course outline

Preliminary, may be subject to change

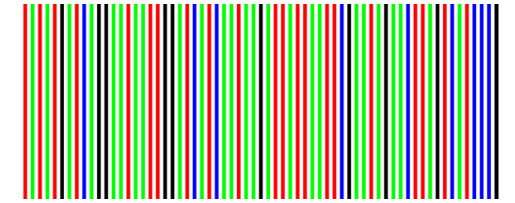
Class	Date	Topics	Script
01	Apr 10	Introduction and setup	01_intro.R
02	Apr 17	Hardy-Weinberg equilibrium	02_hwe.R
03	Apr 24	<i>F</i> -statistics and population structure I	03_popst.R
–	May 01	Labor Day	
04	May 08	Genome sequencing and assembly	04_asm.sh
–	May 15	Himmelfahrt Break	
05	May 22	Variant calling and SNPs	05_vcal.sh
06	May 29	Population genomics and genetic diversity	06_gdiv.sh
07	Jun 05	Population structure II	07_clust.sh
08	Jun 12	Selection	08_sel.R
–	Jun 19	Student presentations (no exercises)	
09	Jun 26	DNA barcoding	09_barcode.sh
10	Jul 03	Metabarcoding / eDNA	10_microb.txt
11	Jul 10	Metabarcoding / eDNA II	11_edna.R

Today's objectives

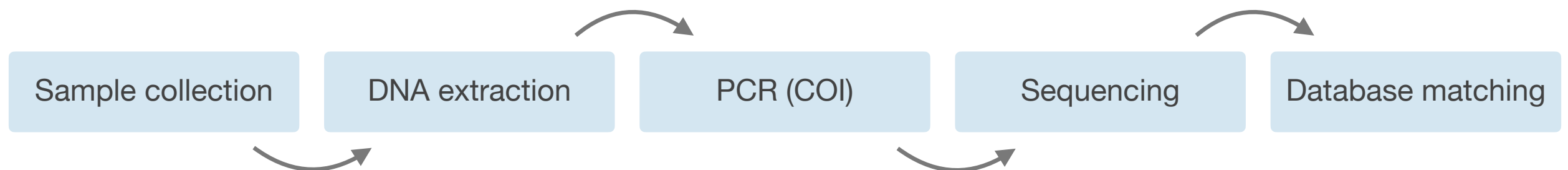
By the end of this session, you will be able to:

- View Sanger sequencing **trace files**, and assess their quality
- Trim and assemble forward and reverse reads into a consensus **DNA barcode** sequence
- Match this barcode to the BOLD database to **identify the species of origin**
- Evaluate identification confidence by comparing p -distances within and between species, and discuss the **barcode gap**

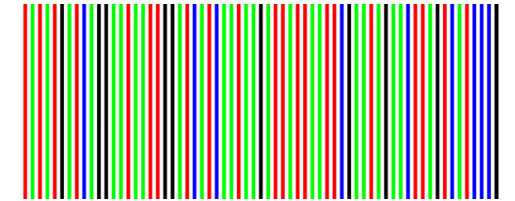
DNA barcoding



- Method of **species identification** using short, standardized gene region (COI, cytb, 16S ...)
- Barcode is compared to **reference sequence database** to find closest match
- **Applications**
 - Identify species from parts or life stages (detect food fraud, wildlife trafficking)
 - Determine species boundaries (complement traditional taxonomy)
 - Analyze whole communities (biodiversity surveys, microbial ecology: metabarcoding)



COI barcode



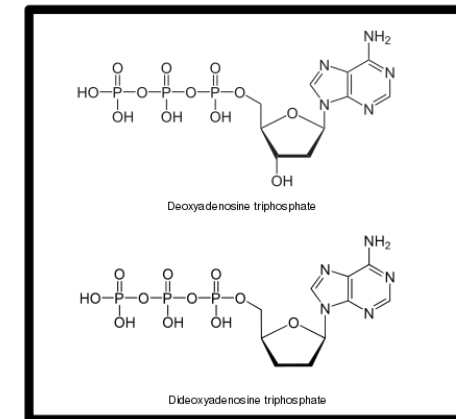
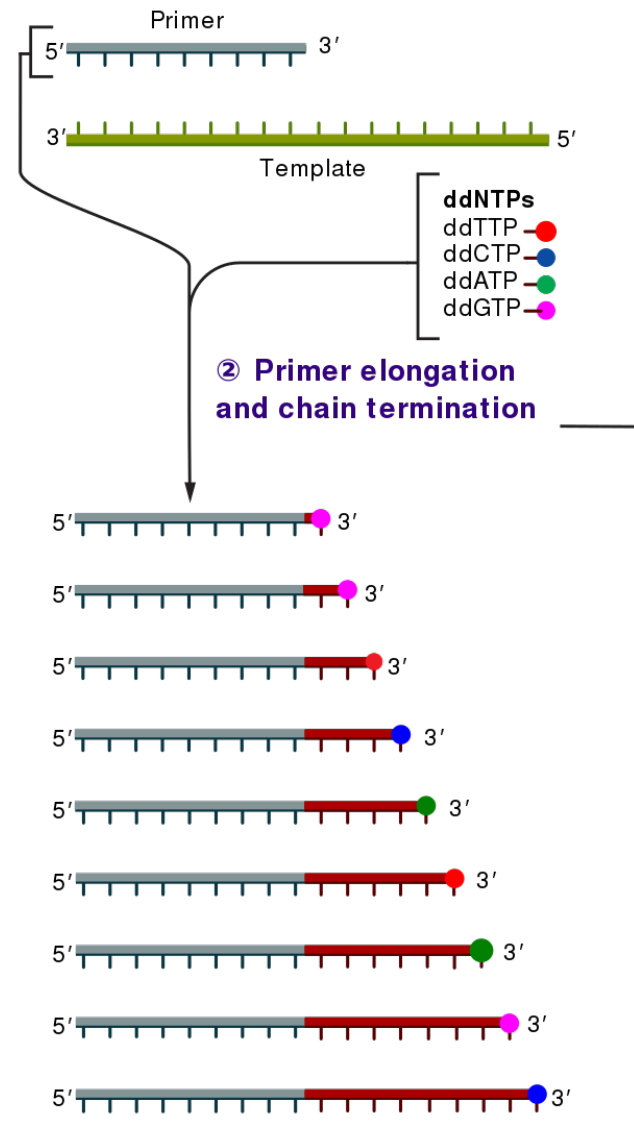
Approx. 650 bp in 5' region of cytochrome c oxidase subunit I (COI)

```
>MN604318.1 Oncorhynchus keta cytochrome c oxidase subunit I gene, complete cds; mitochondrial
GTGGCAATCACACGATGATTCTTCTCAACCAACCACAAAGACATTGGCACCTCTATTTAGTATTTGGTGCCTGAGCCGGGATAGTAGGCACCGCCCTG
AGCCTACTAATTTCGGGCAGAACTAAGCCAGCCAGGCGCTCTTCTAGGGGATGACCAGATCTACAATGTAATCGTTACAGCCCATGCCTTCGTTATAATT
TTCTTTATAGTCATACCAATTATAATCGGAGGCTTTGGAACTGATTAATCCCCCTAATGATCGGGGCACCAGATATAGCATTCCCACGAATAAATAAC
ATAAGCTTCTGACTCCTACCTCCGTCCTTCCTCCTCCTCTTCTTCATCTGGAGTTGAAGCCGGCGCTGGTACCGGGTGGACAGTTTATCCCCCTCTA
GCCGGAACCTTGCCACGCAGGAGCATCTGTCGACTTAACCATCTTCTCCCTCCATTTAGCTGGAATCTCCTCAATTTTGGGGGCCATTAATTTTATT
ACGACCATTATCAACATAAAACCCCCAGCTATTTCTCAGTACCAAACCCCGCTTTTTGTCTGAGCTGTACTAATCACTGCTGTACTTCTACTATTATCA
CTCCCCGTTCTGGCAGCAGGTATTACTATGTTGCTCACAGATCGAAATTTAAACACCACTTTCTTTGACCCGGCGGGTGGCGGAGATCCAATTTTATAC
CAACACCTCTTTTGATTCTTCGGTCACCCAGAGGTCTATATTCTGATCCTCCCAGGCTTTGGTATAATTTACATATCGTTGCATATTACTCTGGTAAG
AAAGAACCTTTTCGGGTACATAGGAATAGTGTGAGCTATAATAGCCATCGGCTTGTTAGGATTTATCGTTTGAGCCCACCACATATTTACTGTCTGGGATG
GACGTGGACACTCGTGCCTACTTTACATCTGCCACCATAATTATCGCTATCCCCACAGGAGTAAAAGTATTTAGCTGACTAGCTACACTGCACGGAGGC
TCGATCAAATGAGAGACACCACTTCTCTGAGCCCTAGGATTTATCTTCCTATTTACAGTGGGCGGATTAACGGGCATCGTCCTTGCTAACTCCTCATT
GACATTGTTTTACATGACACTTATTACGTAGTCGCCCATTTCCACTACGTACTCTCAATAGGAGCTGTATTTGCCATTATGGGCGCTTTCGTACACTGA
TTCCCCCTATTCACAGGGTACACCCTTCACAGCACATGAACCAAATCCATTTTGAATTATATTTATCGGTGTAAATTTAACCTTTTTCCCACAGCAT
TTCCTAGGCCTCGCAGGGATACCACGACGGTACTCTGACTACCCGGACGCCTACACGCTATGAAACACTGTATCCTCAATCGGATCCCTTGTCTCCTTA
GTAGCTGTAATTATGTTCTTATTTATTCTTTGAGAGGCTTTTGCTGCCAAACGAGAAGTAGCATCAATCGAAATAACTTCAACAAACGTAGAATGACTA
CACGGATGCCCCCACCCTACCACACATTTCGAGGAACCAGCATTTGTCCAAGTACGAACGTACTAA
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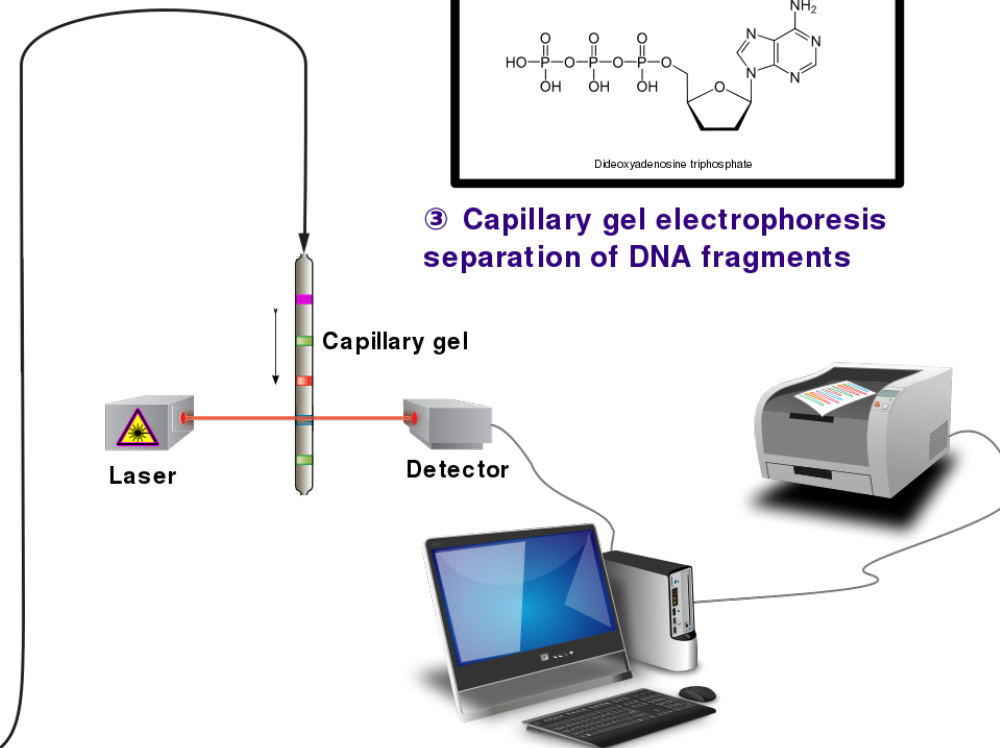
Sanger sequencing

① Reaction mixture

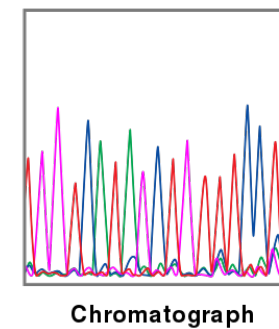
- ▶ Primer and DNA template ▶ DNA polymerase
- ▶ ddNTPs with flouorochromes dNTPs (dATP, dCTP, dGTP, and dTTP)



③ Capillary gel electrophoresis separation of DNA fragments



④ Laser detection of flouorochromes and computational sequence analysis



Estevezj, CC BY-SA 3.0

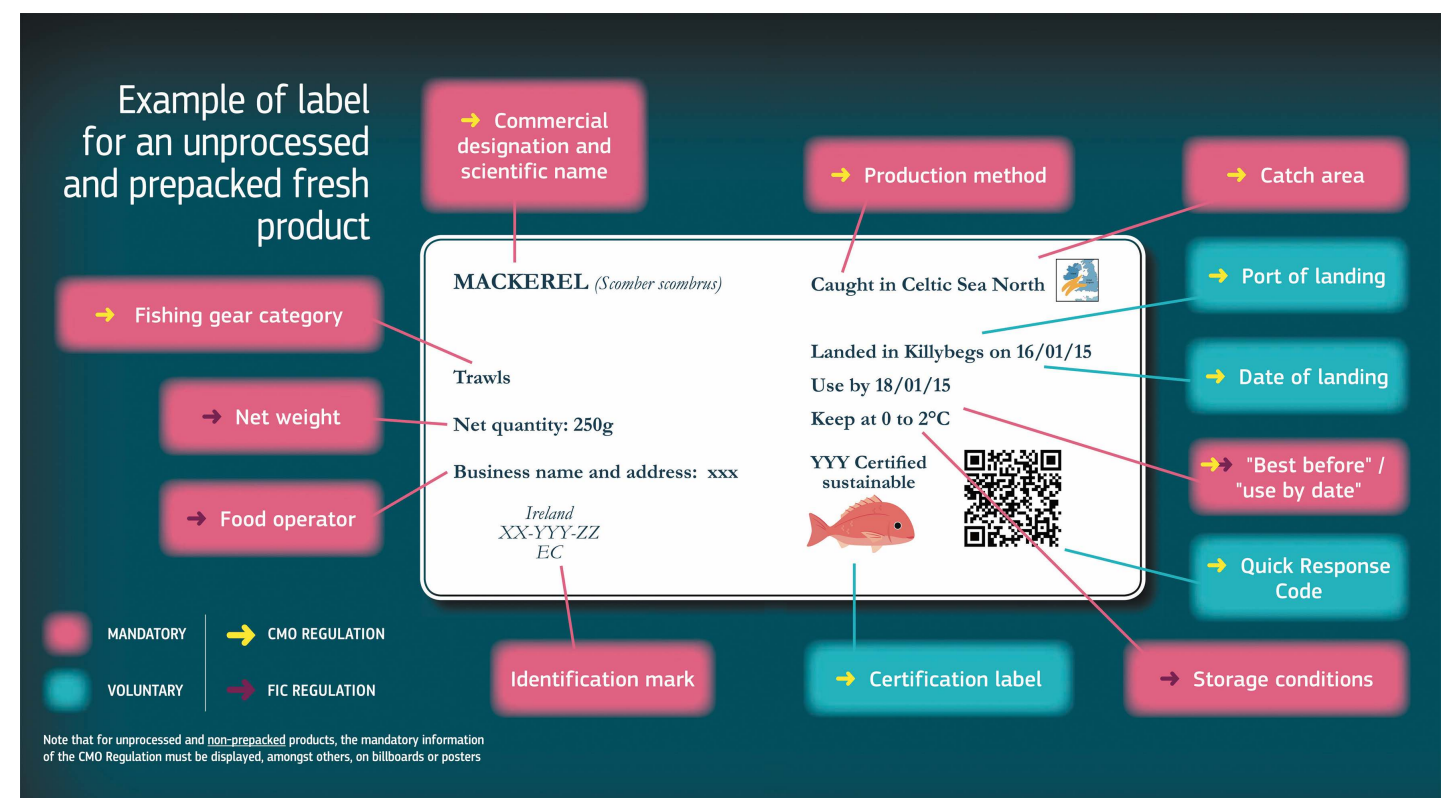
#fischdetektive

Where does our seafood come from, and is it labeled correctly?

Citizen science project
at GEOMAR in 2017 with over
700 participants (10–14 years)

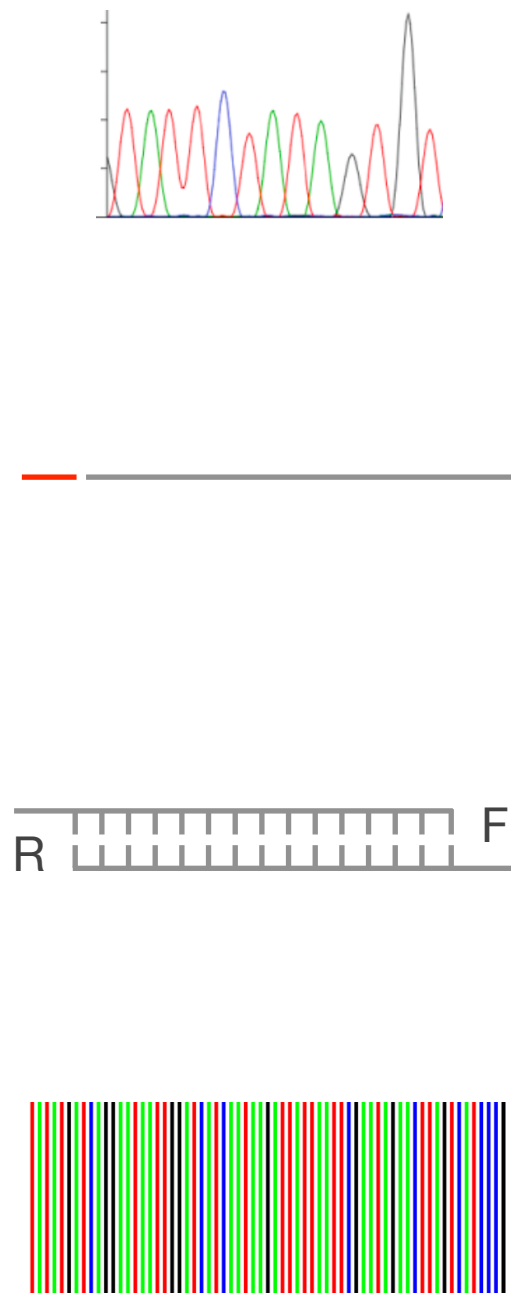


Thorsten Reusch, GEOMAR



Sanger read processing

Exercise 1



Trace file (F + R)



Fasta file (F + R)



Alignment



Consensus

Basecalling

Trimming

Reverse complement R
Align F and R

Assemble consensus sequence

Sequence alignment

Exercise 1

G A T G T T C G A A

G A T C G A A

G A C C T C G T



G A T G T T C G A A

G A T C - - - G A A

G A C C - T C G - T

Alignment

arranges nucleotide or amino acid sequences
so that the number of mismatches and gaps are minimized

- Accomplished by introducing gaps (-), which represent insertions or deletions (**indels**) / account for sequence length differences due to mutations over time
- **Reveals evolutionary or functional relationships** between sequences (e.g. homology)
- Key to variant calling, sequence assembly, species identification and phylogenetics

- **Clusters of similar barcodes**
- Clustering is based on genetic similarity, independent of formal taxonomy
- Represent operational taxonomic units (OTUs) that often correspond to species
- Enable species identification and biodiversity assessments

Exercise 2

Proportion of nucleotides at which two sequences differ

```
AAGCCAGCCAGGCGCTCTTCTAGGGGATGACCAGATCTACAATGTAATCG    # 50 positions total
AAGTCAACCTGGTGCACCTTCTTGGTGATGATCAAATTTATAATGTGATCG
***.**.**.**.***** ** ** *.**.*.***.*****.*****  # 13 differences
```

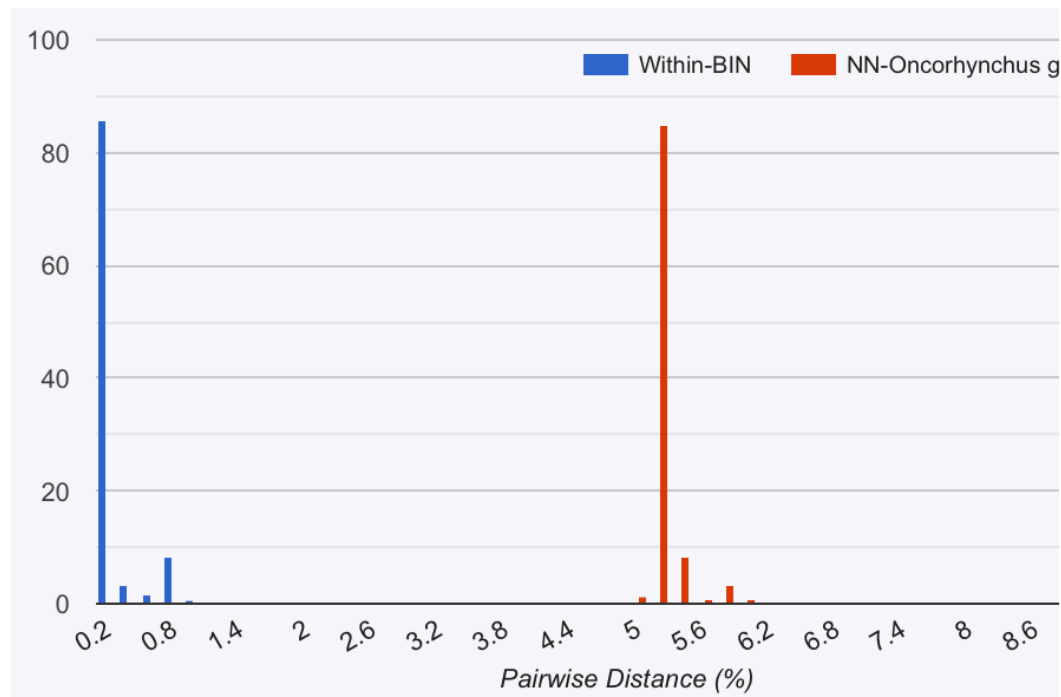
$$p = 13 / 50 = 0.26$$

Does not correct for multiple substitutions (repeated mutations of the same site)
or transition / transversion bias (as K2P distance, Kimura 1980)

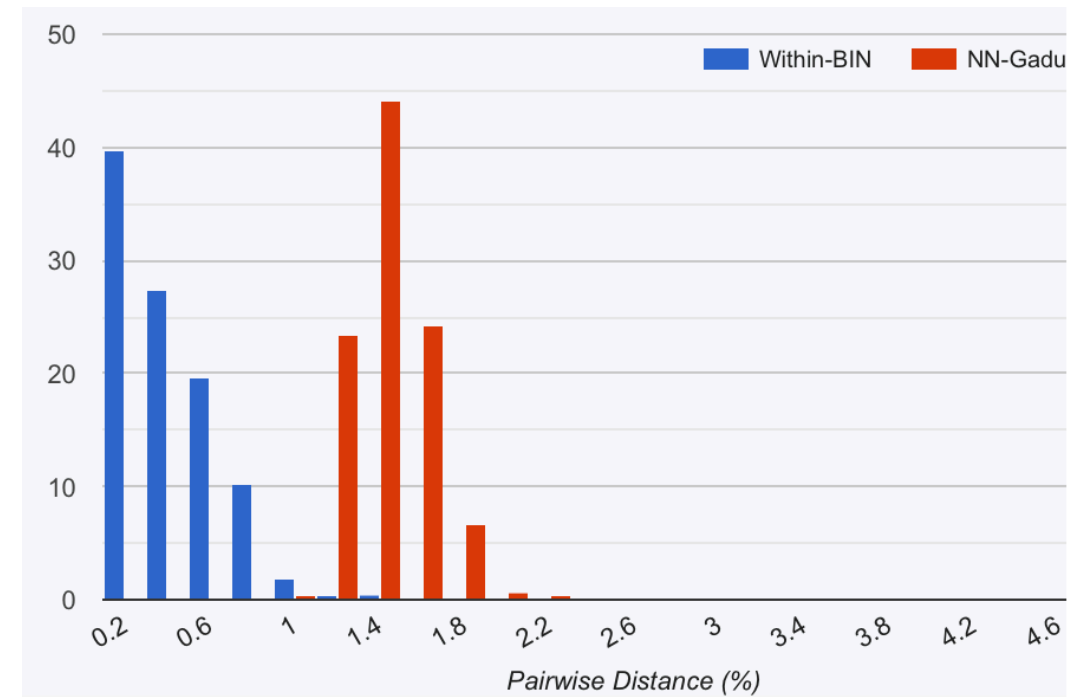
Barcode gap

Comparing genetic distances within BIN
and to nearest neighbor (NN) BIN

Comparison	Typical p -distance (COI)
Within species	< 2 %
Between species	> 2–3 %
Between genera	10–20 %



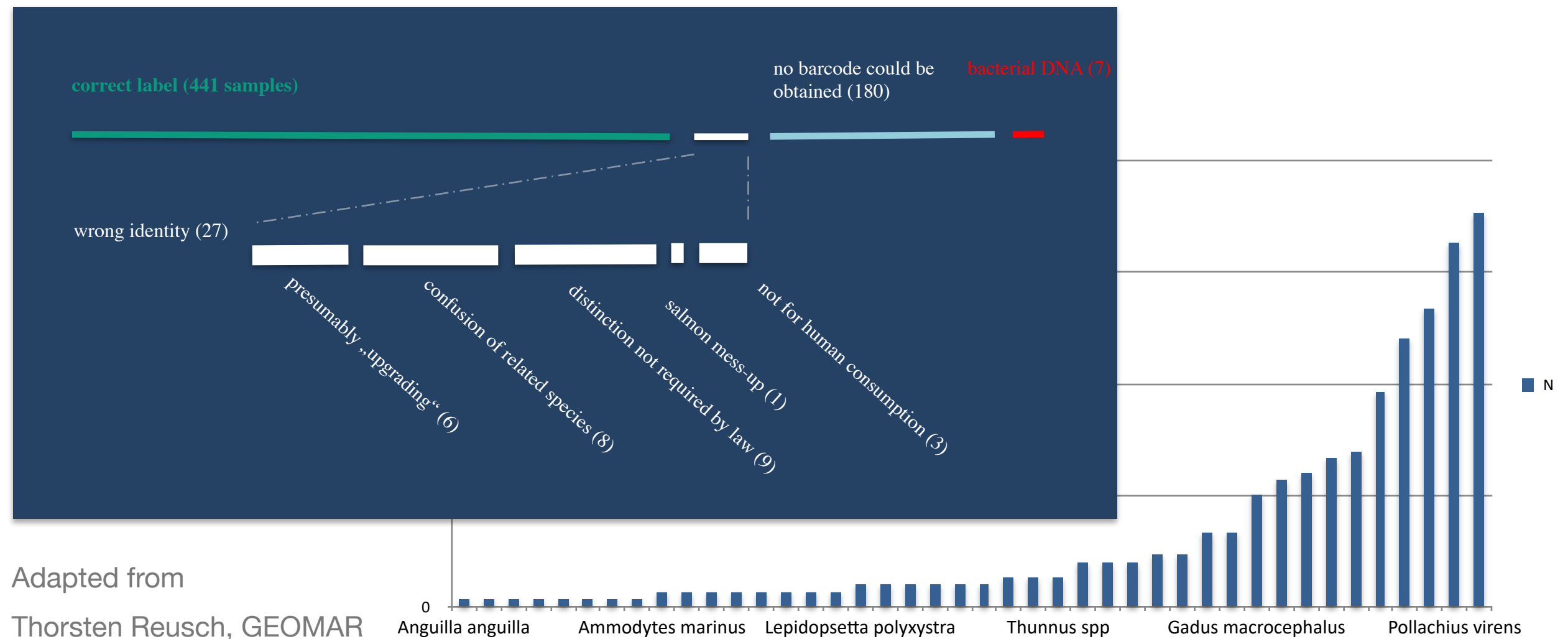
Large barcode gap
Oncorhynchus keta



Small barcode gap
Gadus chalcogrammus

#fischdetektive results

Mislabeling seems to be only a moderate problem in Germany in frozen and fresh fish
(but may be higher in Sushi-grade fish and processed fish products)



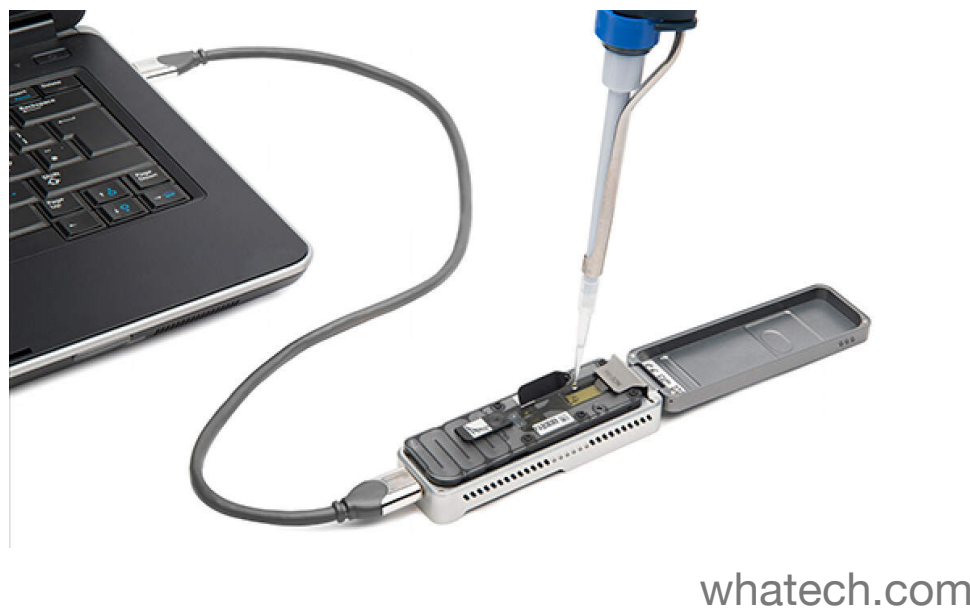
Portable 3rd gen sequencing



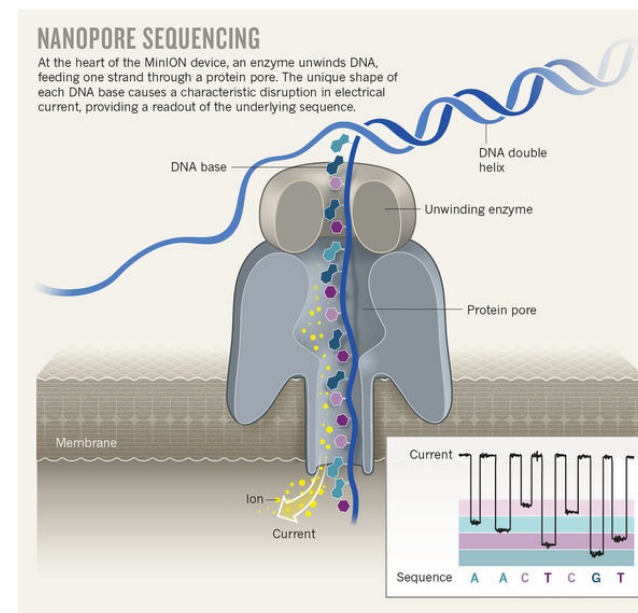
Cretus Joseph Mtonga

PhD project: Build database of the fish fauna in the Western Indian Ocean to enable biodiversity surveys / fisheries monitoring using eDNA

Oxford Nanotechnologies MinION



whatech.com



blogs.nature.com

DNA barcoding

Summary

[illegible]

- **High-quality consensus barcodes** can be obtained from Sanger reads through careful quality trimming and forward-reverse assembly
- Matching to a reference database such as **BOLD enables species assignment**, but confidence depends on sequence quality, database completeness, and taxonomic group in question
- Identification confidence can be assessed by examining the **barcode gap** — the separation between intraspecific variation and distance to the nearest-neighbor species

Please help to [evaluate this course](#)

June 26 – July 17

